Data Analytics Capstone Case Study Report

Recidivism Risk Forecasting & Rehabilitation Strategy

Analyst: Sydney Varner
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Dataset Focus: NIJ Recidivism Forecasting Challenge & Georgia Department of Corrections (GDC) Data

1. Ask Phase

Business Task

The primary objective is to develop a predictive analytics model and evaluation framework to forecast the likelihood of parolee recidivism over 1-year, 2-year, and 3-year timeframes following release. By identifying key risk factors and early warning indicators, the Georgia Department of Corrections (GDC) can optimize resource allocation, tailor community supervision intensity, and provide targeted rehabilitation services to lower overall reoffending rates while advancing public safety and fair treatment.

Key Stakeholders

Stakeholder Group	Primary Interest & Role
GDC Executive Leadership	Strategic decision-making, policy formulation, budget allocation, and state-wide recidivism reduction.
Parole & Supervision Officers	Actionable risk scoring for individualized case management, supervision intensity adjustments, and intervention tracking.

Stakeholder Group	Primary Interest & Role
Rehabilitation Service Providers	Designing targeted behavioral health, vocational training, and educational programs based on specific risk drivers.
Policy Makers & Public Safety Advocates	Ensuring equitable, transparent model outcomes and evaluating the broader societal and economic impacts of correctional programs.

Business Metrics & Success Criteria

- **Model Accuracy & Precision:** Brier Score and Area Under the ROC Curve (AUC-ROC) across 1-, 2-, and 3-year evaluation windows.
- **Recidivism Reduction Rate:** Percentage decrease in reoffending rates among high-risk cohorts receiving targeted interventions.
- **Resource Optimization Index:** Efficiency ratio comparing successful intervention outcomes against supervision costs.
- **Algorithmic Fairness:** Demographic parity in error rates across subgroups to ensure non-discriminatory predictions.

Guiding Business Questions

- Which historical attributes (demographic, offense type, prior record) serve as the strongest predictors of reoffending?
- How does recidivism risk fluctuate across the three post-release time horizons (Year 1 vs. Year 2 vs. Year 3)?
- What post-release dynamic behaviors (e.g., drug test compliance, employment status) provide early warning signals of potential recidivism?
- How can predictive outputs be formatted into actionable decision-support tools for field officers?
- Which rehabilitation programs (e.g., Cognitive Behavioral Therapy, Mental Health/Substance Abuse treatment) and what dosage of program attendance are most effective at reducing 3-year recidivism rates in Georgia?

2. Prepare Phase

Dataset Overview & Organization

The analysis utilizes administrative tracking records from the **Georgia Department of Corrections (GDC)** published under the **NIJ Recidivism Forecasting Challenge**. The dataset tracks parolees released from Georgia state prisons over a comprehensive multi-year observation period.

Fiscal Year (FY) Reconviction Rates (2012–2023 Releases)
Calendar Year (CY) Reconviction Rates (2011–2022 Releases)
NIJ_s_Recidivism_Challenge_Full_Dataset.csv
Reentry and Cognitive Programming.pdf
GDC Substance Use Reentry Services

Data Schema & Categories

The data is structured tabularly at the individual level (one record per parolee) across several core functional domains:

- **Demographics & Baseline Attributes:** Age at release, gender, race, and gang affiliation status.
- **Criminal History:** Prior felony convictions, violent crime history, property/drug offense counts, and prior revocations.
- **Incarceration & Supervision Parameters:** Time served, initial risk assessment score, supervision district, and initial housing condition.
- **Dynamic Post-Release Variables:** Monthly drug test results (THC, Cocaine, Meth, Opiates), supervision violations, and employment quarterly percentages.
- **Target Outcome Variables:** Binary indicators (`0` = No, `1` = Yes) for recidivism within Year 1, Year 2, and Year 3.

Data Credibility (ROCCC Framework)

ROCCC Dimension	Rating	Assessment
Reliable	**High**	Data is officially captured and validated by state law enforcement agencies and GDC administrative systems.
Original	**High**	Primary administrative source dataset compiled directly by GDC and managed by the National Institute of Justice.
Comprehensive	**High**	Includes over 25,000 individuals with rich longitudinal variables tracking pre-release and post-release factors.
Current	**Medium**	Represents historical release cohorts spanning a 3-year tracking window; ideal for longitudinal trend modeling.
Cited	**High**	Publicly cited and sponsored by the U.S. Department of Justice (DOJ) and National Institute of Justice (NIJ).

Data Integrity, Privacy & Security

- **Anonymization & Privacy:** All personally identifiable information (PII)—including names, SSNs, and exact dates—has been stripped and replaced with unique synthetic identification numbers by NIJ/GDC.
- **Data Validation Checks:** Data structure checks were implemented to handle missing values, address implicit class imbalances, and filter out inconsistent temporal entries.
- **Ethical Safeguards:** Demographic attributes are analyzed to detect and mitigate potential systemic bias, ensuring predictive models prioritize behavioral indicators over static demographic markers.
- **Data Handling Compliance:** All working files are processed in secure environment storage with strict compliance regarding research ethics and data protection standards.

3. Process Phase (Data Cleaning & Manipulation)

1. Tool Selection & Environment

- **Platform:** Google Colab / Python 3.
    - Colab:
      Recidivism Risk Forecasting & Rehabilitation Strategy in Georgia ...
- **Core Libraries:** `pandas` for data structuring, grouping, and aggregation; `numpy` for vectorized conditional logic (`np.select`); and `openpyxl` / `pd.ExcelWriter` for multi-tab spreadsheet generation.

Data Hygiene, Anomaly Detection & Cleaning

A rigorous data hygiene workflow was executed in Python (Pandas/NumPy) and verified with SQL filtering to validate data structure before analytical modeling:

- **Systematic Anomaly Auditing:** Checked every column for blanks and nulls using `.isna()` and SQL filtering. This isolated missingness to dynamic behavioral fields rather than systemic record corruption.
- **Preservation of High-Risk Populations:** Avoided standard `dropna()` filtering to prevent dropping over 5,000 individuals, ensuring all high-risk non-compliant parolees remained in the cohort.

2. Step-by-Step Data Transformations

1. **Dose-Response Aggregation:** Formatted `Program_Attendances` into discrete dosage tiers (0 to 10+ sessions) and calculated the mean recidivism rate and population headcount.
2. **Continuous Variable Binning:** Used `pd.cut()` to discretize continuous employment percentages (`Percent_Days_Employed`) into five policy-relevant tiers: `0%`, `1–25%`, `26–50%`, `51–75%`, and `76–100%`.
3. **Vectorized 3-Tier Substance Use Segmentation:** Implemented `np.select()` to classify parolees into three mutually exclusive categories:
    - `Tested - Clean`: Tested with 0 positive screens.
    - `Tested - Positive`: Tested with at least one positive screen across THC, Cocaine, Meth, or Other.
    - `Missing / Un-tested`: Missing drug test records entirely (non-reporting / absconding proxy).
4. **Systemic Facility Normalization:** Cleaned GDC facility category strings (`'Transition centers'`, `'County CI'`, `'State prison, IBCs'`, `'Private prisons'`, `'All inmate facilities **'`), coerced multi-year release figures into numeric formats, and computed longitudinal row-wise means, 10-year minimums, and 10-year maximums.
5. **Multi-Tab Excel Export:** Structured each summary table as a dedicated DataFrame and compiled them into an automated, multi-tab Excel workbook (`Recidivism_Case_Study_Summary.xlsx`) for chart generation in Google Sheets.

Colab Verification Artifact:
Link to Exported Summary File:
`Recidivism_Case_Study_Summary.xlsx` *(Tabs:*

Program_Attendance, *Employment_Tiers*, *Substance_Use*, *Facility_Comparison*)

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4. Analyze Phase

1. Key Surprises Discovered in the Data

1. **Non-Reporting Outweighs Relapse as a Risk Signal:** Individuals who missed drug tests entirely reoffended at **67.6%**, whereas individuals who tested positive for drugs reoffended at **62.2%**. This indicates that failing to maintain contact with supervision officers is a more acute predictor of imminent rearrest than clinical drug relapse.
2. **The "Early Compliance Mandate" Bump:** Parolees attending only 1–3 program sessions exhibited higher recidivism (~66%) than those attending 0 sessions (59.1%). This reflects administrative mandates where high-risk individuals are ordered to participate in programs but drop out before receiving a therapeutic benefit.
3. **The 10+ Session Protective Threshold:** Achieving **10 or more attendances** yields a dramatic decline in recidivism down to **47.1%** (a 12.0 percentage point reduction compared to non-attenders). Dosage and retention, rather than mere enrollment, drive rehabilitation success.
4. **Transition Centers Drastically Outperform Custodial Prisons:** GDC Transition Centers recorded a **12.3% reconviction rate in 2022** and a **17.2% multi-year average**, compared to **31.1%** for the state baseline and **33.8%** for standard state prisons. Transition centers cut reconviction rates by more than half across an entire decade of releases.

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2. Summary Tables of Core Calculations

Table 1: Substance Testing Status vs. 3-Year Recidivism Rate

```
None
     Drug_Test_Status  recidivism_rate  headcount
0  Missing / Un-tested             67.6       5172
1       Tested - Clean             50.2      12024
2    Tested - Positive             62.2       8639
```

Drug Test Status	Headcount	3-Year Recidivism Rate (%)	Strategic Implication
Tested - Clean	12,284	**50.2%**	Demonstrates baseline compliance and effective community reintegration.
Tested - Positive	8,379	**62.2%**	Active substance use elevates risk; indicates immediate need for clinical relapse care.
Missing / Un-tested	5,172	**67.6%**	Highest-risk cohort; signals non-reporting, absconding, and loss of supervision contact.

Table 2: Employment Bracket vs. 3-Year Recidivism Rate

```
None
Percent_Days_Employed  False  True
Employment_Exempt
False                  22015    384
True                    3358     78
  Employment_Group  recidivism_rate
0               0%             67.2
1            1–25%             79.8
2           26–50%             73.6
3           51–75%             57.0
```

```
4              76–100%                43.6
```

Employment Group	Headcount	3-Year Recidivism Rate (%)	Strategic Implication
0% Days Employed	6,430	**67.2%**	Complete unemployment is the single largest socioeconomic risk factor.
1–25% Days Employed	2,050	**79.8%**	Early job instability creates acute reoffending vulnerability.
26–50% Days Employed	2,354	**73.6%**	Partial employment begins to mitigate risk but remains fragile.
51–75% Days Employed	2,904	**57.0%**	Crossing the 50% employment threshold delivers substantial risk reduction.
76–100% Days Employed	11,635	**43.6%**	Sustained employment reduces recidivism by **23.6 percentage points**.

Table 3: Institutional Facility Reconviction Comparison (GDC CY 2011–2022)

None

```
          Facility Type  2022  Multi_Year_Mean  10_Year_Min
10_Year_Max
      Transition centers  12.3             17.2         12.2
20.7
All inmate facilities **  31.1             27.1         23.8
31.1
```

```
         Private prisons  32.3             29.3         25.3
32.5
               County CI  32.8             27.7         22.4
32.8
      State prison, IBCs  33.8             28.1         25.1
33.8
```

Facility Type	2022 Rate (%)	Multi-Year Mean (%)	10-Year Min (%)	10-Year Max (%)	Performance Relative to Baseline
🌟 **Transition centers**	**12.3%**	**17.2%**	12.2%	20.7%	**~19% below baseline** (Exceptional reform)
🏢 **All inmate facilities ****	**31.1%**	**27.1%**	23.8%	31.1%	State Baseline Average
🔒 **Private prisons**	**32.3%**	**29.3%**	25.3%	32.5%	~2% above baseline
🛡️ **County CI**	**32.8%**	**27.7%**	22.4%	32.8%	~1% above baseline
⛓️ **State prison, IBCs**	**33.8%**	**28.1%**	25.1%	33.8%	~3% above baseline (Highest risk)

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3. Synthesized Findings & Client Recommendations

Connecting individual behavior patterns with macro facility structures provides an airtight, actionable narrative for correctional leadership:

1. **Workforce Reintegration is the Primary Shield:** Moving an individual into stable employment (>75% of days) reduces the likelihood of reoffending from **67.2% down to 43.6%**. Reentry programs should treat vocational training, resume support, and employer tax credit partnerships as front-line public safety tools.

2. **Shift Focus from Program Enrollment to Retention:** Low program dosage (1–3 sessions) does not reduce reoffending. Therapeutic benefits only materialize when individuals achieve **10+ sessions (47.1% recidivism)**. Case managers should be incentivized on completion and attendance milestones rather than initial referrals.
3. **Modernize Supervision to Intervene on Non-Reporting:** Because missing drug tests carry higher risk than testing positive (67.6% vs. 62.2%), missed tests should trigger supportive field interventions and outreach rather than immediate punitive revocation.
4. **Reallocate Capital to the Transition Center Model:** Transition Centers consistently cut recidivism by more than half compared to standard prisons (12.3% vs. 31.1%). Transition Centers succeed because they provide the exact structural conditions validated by the micro-data: mandatory employment placement, structured community accountability, and sustained program continuity. Correctional departments should reallocate funding away from custodial private beds and into community transition facilities.